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Chapter 22
Staying in Action: Risk of Ruin

Throughout this book, we have focused almost exclusively on expectation as a guide to proper play. Our assumption, stated at the outset, was that all players were adequately bankrolled for the games and that their only concern was to maximize EV. For this chapter, however, we suspend this assumption and consider instead the important question, "What does it mean to be adequately bankrolled for a game?"

Of course, there is no one single answer to this question; all players have different tolerances for risk. For some people, the risk of losing their entire poker bankroll is unacceptable: for others, playing poker may be just like a startup business, where a relatively high chance of failing is acceptable in pursuit of profit maximization. We can, however, apply quantitative methods to the problem by making assumptions, figuring out the implications of those assumptions, and building a model. One model that we use frequently in assessing bankroll requirements is called the ***Risk of Ruin*** (RoR) model.

A substantial part of the following was originally worked out by Bill Chen and Tom Weideman on rec.gambling.poker.

This model assumes the following:

- We repeatedly play a fixed game with a fixed distribution of outcomes **X,**
- Each trial is independent and randomly selected from **X.**
- We have a starting bankroll *b.*
- The bankroll *b* is modified by the outcome of each trial.

We play the game indefinitely, or until the bankroll is reduced to zero.

For the assumptions outlined above, there are always only two possible outcomes for an infinite number of trials. The first possibility is that the bankroll grows without bound; that is, we win more and more and our long-term win rate approaches our win rate in the game. The second is that we lose the entire bankroll at some point. The value we are interested in obtaining is the probability that the starting bankroll will be lost under these conditions. We call this function the ***risk of ruin function,*** $R_X(b)$. We often shorten this to $R(b)$ when dealing with a single game and its corresponding distribution.

This function has certain properties that will allow us to describe its behavior and eventually find its value for a particular game.

Property 1.
If there is no probability of a negative outcome, then $R(b)=0$ for all b.

If we never have a negative outcome as a result of playing the game, then there is no chance of losing the entire bankroll.

Property 2.
If the game has negative expectation, then $R(b) = 1$ for all b.

This property is a result of the fact that the game will be played indefinitely. If the game has negative expectation, then the expected amount lost will grow without bound as the number of

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trials goes to infinity. Contrast this with the situation where expectation is positive. Here the amount won will grow without bound except in the case where we lose the entire bankroll first.

Property 3.

If a game has any possible negative outcomes in **X,** then $R(b) > 0$ for all b.

This property is easily justified as follows. Let's say that the probability of a certain negative outcome X_L is p, and the value of that negative outcome is v. If we have a bankroll B, then we can lose the entire bankroll immediately by having the outcome X_L occur B/v times in a row. This probability is simply $p^{B/v}$. This probability might be incredibly small, but as long as there exist any possible negative outcomes, $R(b) > 0$.

Property 4.

If a game has positive expectation and bounded negative outcomes, $R(b) < 1$ for all b.

Suppose X has positive expectation. Then let X_n be a sequence of independent trials. Define the cumulative result:

$c_n = \sum_{j=1}^{n} x_j$ then we know that with probability 1: $< X >= \lim_{n\to\infty} \frac{c_n}{n} > 0$

Suppose that $R(b)=1$. Then also with (ruin) probability 1, we know that there will be a cumulative result that is negative for some C_n. Furthermore for each C_n we know there will be an m>n such that $C_m<C_n$. This means we have a subsequence:

$0> C_{n1}> C_{n2}> C_{n3}>...$

This contradicts the original assumption that X has positive expectation.

Property 5.

The risk of ruin function for the sum of two bankrolls is equal to the product of the individual risk of ruin values for each bankroll.

$$R(a + b) = R(a)R(b) \qquad \textbf{(22.1)}$$

This property forms the basis for the derivation of the RoR function. If we have a bankroll of $500, and some risk of ruin given that size bankroll, what is the chance that we go broke with $1000? After we have lost the first $500, which occurs with some specified probability, we now have another $500 bankroll with which to play. Since losing the bankrolls are independent events, the joint probability of losing two $500 bankrolls (equivalent to losing one $1000 bankroll) is simply the probability of losing a $500 bankroll squared.

There are some definitional issues with this statement; for example, say we have a game that consists of a $100 wager, but we only have $50. In this sense, we have been "ruined" already. However, we take the position that from an RoR standpoint, since we can put two $50 bankrolls together to form a $100 bankroll to play the game, $50 has a lower risk of ruin than $0 for this game. When we consider poker games, this distinction effectively never comes up, so we simply neglect it.

Example 22.1

Let us look at a simple example. Suppose we play the following game. We roll a die, and if the the comes up 1 or 2 we lose $100. If the die comes up 3-6, we win $100. This is a great game! We win more than $33 per toss. But suppose we only have $100. What is our chance of

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eventually losing this $100 if we play the game indefinitely? Using the properties above, we can actually solve this problem.

On the first roll, we have a $^1/_3$ chance of going broke, and a $^2/_3$ chance of growing our bankroll to $200. Suppose we call $100 "one unit." Then the risk of ruin of a bankroll of $100 is R(l), and the risk of ruin of a bankroll of $200 is *R(2).*

$R(1) = {}^1/_3 + {}^2/_3\, R(2)$

We know From Property 5 above that $R(a + b) = R(a) + R(b)$, so:

$R(2) = R(1)R(1)$

Substituting and solving:

$R(1) = {}^1/_3 + {}^2/_3\, R(1)^2$
$2R^2 - 3R + 1 = 0$
$(2R - 1)(R - 1) = 0$

This gives us two possible risk of ruin values, $^1/_2$ and 1. We know that this game has positive expectation and therefore by Property 4 above, the risk of ruin must be less than one. Hence, *R(1)* for this game is $^1/_2$.

The reader can actually verify this by trials. Roll dice and play the game until either losing the initial $100 stake or winning $1000. If you repeat this process enough times, the results will converge on $^1/_2$,

Remming to Equation 22.1:

$R(a+ b)=R(a)R(b)$

We can do some algebraic manipulation which will help us to expose an important property *of R(x).*

First, we take the natural logarithm of each side:

$\ln R(a + b) = \ln R(a) + \ln R(b)$

If we make up a function $f(x) = \ln R(x)$, then we have:

$f(a + b) = f(a) + f(b)$

This relationship shows that f is linear. We can see, for example:

$f(1+1) = f(1) + f(1)$
$f(2) = 2f(1)$
$f(2+1) = f(2) + f(1)$
$f(3) = 3f(1)$

$f(n) = nf(1)$

...and so on.

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So f (x) has the form f (x) = -αx, where α is a constant. (This is not to be confused with the α of Equation 11.1, the ratio of bluffs to value bets.) The risk of rum for every game has this form, and we call α the ***risk of ruin constant***. This causes α to be positive, α is actually equal to the natural logarithm of the risk of ruin of a bankroll of 1 unit. Because we know that R(x) is between 0 and 1 for all x, we add the negative sign by convention, as ln x is negative for x between 0 and 1.

a = ln (R(l))

α is a constant that tells us how much one unit of bankroll is worth in terms of risk of ruin. Suppose that we have one unit of bankroll. Then our risk of ruin is $e^{-\alpha}$. *(e* is a common mathematical constant - approximately 2.71 - that appears frequently throughout Part TV). We use the notation exp(x) and e^x interchangeably for ease of reading. Suppose that we have two units of bankroll. Then our risk of ruin is $e^{-\alpha}$ times $e^{-\alpha}$, or $e^{-2\alpha}$. And our risk of ruin for N units of bankroll is just $e^{-N\alpha}$.

So we have:

$$In\ R(x) = -\alpha x$$
$$R(x) = e^{-\alpha x}$$

So the form of the risk of ruin function is an exponential depending on the bankroll and some constant that is a function of the structure of the distribution of outcomes of the game.

Let's reconsider the example we had. Based on what we showed previously, we know that the risk of ruin for the 1 unit bankroll is $^1/_2$. The game where the bankroll is $100 and the outcomes are +$100 and $100 is equivalent to the game where the bankroll is 1 unit and the outcomes are +1 units and -1 units.

$$R(b) = e^{-\alpha b}$$
$$R(1) = {}^1/_2$$
$${}^1/_2 = e^{-\alpha}$$
$$-\alpha = ln\ {}^1/_2$$
$$\alpha = ln\ 2$$

There is one more important property of the risk of ruin function.

Property 6.

For one play, $R(B) = \langle R(B + x) \rangle$, where x is a random variable selected from the outcome distribution **X**.

This property states that the risk of ruin for a bankroll B is the expected value of the risk of ruin of the bankroll after one additional trial. This is because the risk of ruin model assumes that you will play the game indefinitely, one dial at a time. So if you have a bankroll *B* with a risk of ruin *R(B)* right now, and you play the game once, *R(B)* is the weighted average of your possible bankrolls after this trial. This may be a little confusing, but an example should help to show its simplicity.

Example 22.2

In our example game, we have the following outcomes:

+1, probability $^2/_3$

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-1, probability $^1/_3$.

What this property says is that for any bankroll,

$R(B) = \langle R(B + x) \rangle$
$R(B) = {}^1/_3 R(B - l) + {}^2/_3 R(B + 1)$

This relationship describes the relationship between risk of rain and the expected outcomes of the game. After one trial, there is some number of discrete possible bankrolls, each corresponding to a particular outcome of the game. The expected value of the risk of ruin of each of those bankrolls must equal the current risk of ruin. This game is repeated an infinite number of times.

Using this, we have:

$R(B) = \langle R(B + x) \rangle$
$e^{-\alpha B} = \langle e^{-\alpha B} e^{-\alpha x} \rangle$

On the right side, $e^{-\alpha B}$ is a constant, so we can pull that outside of the expectation brackets:

$e^{-\alpha B} = e^{-\alpha B} \langle e^{-\alpha x} \rangle$

$1 = \langle e^{-\alpha x} \rangle$ **(22.2)**

We can use this important relationship to find the risk of ruin constant for simple risk of rain problems directly, such as the example we have been working on:

$^1/_3 e^{\alpha} + {}^2/_3 e^{-\alpha} = 1$
$e^{\alpha} + 2e^{-\alpha} = 3$
$1 + 2e^{-2\alpha} = 3e^{-\alpha}$

Letting $x = e^{-\alpha}$

$1 + 2x^2 = 3x$
$2x^2 - 3x + 1 = 0$
$(2x - 1)(x - 1) = 0$
$x = {}^1/_2$ or $x = 1$

$e^{-\alpha} = {}^1/_2$ or $e^{-\alpha} = 1$

When we solve equation 22.2 for any game, one of the solutions is always $\alpha = 0$. For games with EV < 0, this is the correct solution to use, as $R(b) = 1$ for these games. However, for games with EV > 0, the answer is the second root, in this case $\alpha = \ln 2$. Again, this matches our previous solutions.

We should take some time here to review what we've done so far. First, we created a set of assumptions about a function called the risk of ruin function, which we called $R(b)$. Then we enumerated some properties of this function, including the extremely important property $R(a + b) = R(a)R(b)$. From this and the other properties of the function, we were able to derive a more direct formula for the risk of ruin of any game, $R(b) = e^{-\alpha b}$, where α is a constant that is a property of the game being studied. However, at this point, we had no reasonable way to find α other than solving for $R(l)$ directly.

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Next, we stated an additional property of the function *R(b)*, which explained the relationship of the *R(b)* to the expected value of risk of ruin after a single trial. Using this relationship, we were able to cancel out the bankroll-dependent portion of the risk of ruin function and show that ***we can find the value of a directly from the parameters of the game, without solving any risk of ruin for any particular case.*** This makes an entire class of problems tractable that would have been difficult under other circumstances. We continue by observing some things about the value of α.

We can make some interesting observations about the behavior of the function $<e^{-\alpha x}>$, which we will call g(a) for now.

First, we know that g(0) = 1.

Second, we can take the partial derivative of g with respect to α to obtain:

$g'(a) = < -x\, e^{-\alpha x} >$. $g(0)$ then $= < -x >$, Since we know that the game has positive expectation, $g'(0)$ is always negative. So at 0, *g* is 1 with a downward slope.

Third, we can find $g''(\alpha) = < -x^2 e^{-\alpha x} >$. This value is always positive.

Fourth, we can see that if there are any negative outcomes in **X**, then *g(x)* goes to infinity as α goes to infinity, because all terms with positive outcomes go to zero while those with negative terms increase without bound.

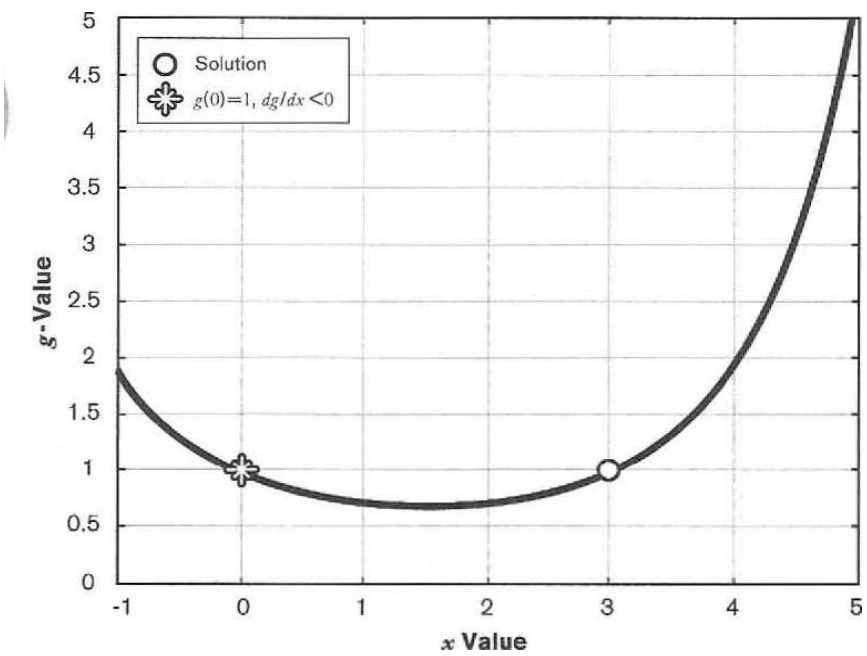

Figure 22.1. A sample of g(x) for a game with positive EV.

Taking all these findings together, the graph of *g* must begin at 1, slope downward but with upward concavity, and eventually Increase without bound as *a* goes to infinity. Therefore a solution to Equation 22.2 exists where *a* > 0, and that solution is unique. This obviates a potential difficulty that could arrive if we had several different *a* values that all satisfied Equation 22.2. Once we find a non-zero value of *a*, we know that it is the correct value to use.

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This suggests, then, a method for solving any risk of ruin problem where the distribution of outcomes is known, which is to solve Equation 22.2 for the expected distribution of outcomes. We consider a more complex and real-world example next.

Example 22.3

Consider the case of a player on a limited bankroll. He is fairly risk-averse, as he has neither another means of support nor any way to raise an additional bankroll. He has decided that $200+15 sit-and-go tournaments are the best mixture of return and variance reduction for him. These tournaments pay out the following:

1st place: $1000
2nd place: $600
3rd place: $400

His expected finishes in the tournaments are as follows:

Finish	**Probability**	**Net**
1st	0.12	$785
2nd	0.13	$385
3rd	0.13	$185
4th-10th	0.62	-$215
Total	**1**	**EV = $35/tournament**

What is his risk of ruin for various bankroll sizes?

The method to use here is to set up Equation 22.1 for this player's outcome distribution.

$$0.12e^{-a(785)} + 0.13e^{-a(385)} + 0.13e^{-a(185)} + 0.62e^{-a(215)} = 1$$

Now this equation is not easily solvable by hand; however, there are many tools, such as Microsoft Excel, that include "solver" utilities that can solve such problems by converging on *a* solution. By putting the above equation into one of these, we have:

$a \approx 0.000632$
$R(b) = exp(-0.000632b)$.

Looking at the results of this for some different possible bankroll sizes:

Bankroll Size	**Risk of Ruin**
500	72.9%
1000	53.2%
2000	28.3%
3000	15.0%
4000	8.0%
5000	4.3%
7500	0.9%
10000	0.2%

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A Word of Caution

Before we proceed any further, let us make particular note of an important assumption that underlies risk of ruin. This assumption is that the player will put all the money won in the game back into the bankroll and let it grow without bound. When we look at risk of ruin values as percentages, the temptation is to view these values as an ongoing and permanent chance that we will go broke given a certain bankroll that generates motley that we skim off the top. We cannot be too clear about this misconception.

The risk of ruin of a bankroll that is maintained at a particular level and from which all the profit above that level is taken out of the bankroll is 1, or a 100% chance.

If any money is taken out of the bankroll even if it is from profit, then the risk of ruin of that bankroll will naturally increase. Risk of ruin takes into account that some of the paths include making enough money initially to sustain a long losing streak in the future. There are many reasonable strategies for extracting money from a bankroll in a manner that allows us to rationally consider risk, but all of them increase the risk of ruin.

Half-Bankrolls

Those readers familiar with the sciences will recognize the structure of the risk of ruin function as being similar to that of other functions, such as radioactive decay. A radioactive substance contains (usually) certain isotopes of an element that are unstable. The unstable isotopes decay in a random way over a period of time, called the half-life. During this amount of time, the random decay process causes half of the radioactive matter inside a sample to decay. However, after two half-fives, the sample is not completely decayed. Instead, during the second period, half of the *remaining* sample decays, leaving $^1/_4$ of the original, and so on. The sample never truly decays completely, but simply approaches zero over a large number of half-lives.

We have analogized the process of risk of ruin to this; both are exponential functions and behave in similar ways. A ***half-bankroll*** is the amount of bankroll required such that the risk of ruin of the bankroll is exactly one-half. Then in the same way, the RoR of two half-bankrolls is $^1/_4$ of three half-bankrolls is $^1/_8$, and so on.
To find a formula for generating half-bankrolls, we simply solve the RoR equation such that the RoR is $^1/_2$:

$e^{-ab} = {}^1/_2$
$-ab = -\ln 2$
$b = (\ln 2)/a$

So the size of a half-bankroll is $(\ln 2)/a$. We can approximate the half-bankroll for the sit-and-go tournament situation described earlier; recall that ln 2 is about 0.693 and a was about 0.00632. So the size of a half-bankroll is approximately $1100. Looking at the table of risk of ruin values above, we see that if we interpolate the risk of ruin for $1100, $2200. $3300, etc, we see that they follow the geometric pattern of 0.5^b, which we expect.

Half-bankrolls are simply a special case of the more general risk of ruin formula, and can be of some use for approximating bankrolls required for various levels of risk of ruin. We often use these as a shorthand for the more complex risk of ruin formula as they can easily be compared across different games. Once we have solved for any particular risk of ruin, we can easily find the half-bankroll and then put together multiples of the half-bankroll for any desired risk of ruin.

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Risk of Ruin for Normally Distributed Outcomes

Next we consider the risk of ruin formula for a game in which the distribution **X** is normal, with a mean μ and a standard deviation σ.

Calling *v*(x) the value function and p(x) the probability density function for that value function, we have the following:

$$< v(x) >= \int_{-\infty}^{\infty} p(x)v(x)dx \qquad \textbf{(22.3)}$$

This identity provides the basis for extending our basic risk of ruin formula to this case where the distribution is normal.

The normal distribution is defined by its characteristic density function:

$$N(x) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right) exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$$

where μ is the mean of the distribution and σ its standard deviation.

We begin with equation 22.2:

$<e^{-\alpha x}> = 1$

Combining this with equation 22.3, we have:

$$\int_{-\infty}^{\infty} e^{-\alpha x} N(x)dx = 1$$

$$\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} \exp(-\alpha x)\, exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)dx = 1$$

$$\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} exp\left(\frac{-[(x-\mu)^2 + 2\alpha\sigma^2 x]}{2\sigma^2}\right)dx = 1$$

Considering briefly only the exponent of *e*, we have:

$$\frac{-(x-\mu)^2 + 2\alpha\sigma^2 x}{2\sigma^2}$$

$$\frac{-(x^2 - 2x\mu + \mu^2) \quad + 2\alpha\sigma^2 x}{2\sigma^2}$$

$$\frac{-(x^2 - 2x\mu + 2\alpha\sigma^2 + \mu^2)}{2\sigma^2}$$

$$\frac{-(x^2 - 2x(\mu - \alpha\sigma^2) + \mu^2)}{2\sigma^2}$$

We can then complete the square of the numerator of this function to obtain:

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$$\frac{-(x^2 - 2x(\mu - \alpha\sigma^2) + (\mu - \alpha\sigma^2)^2 + \mu^2 - (\mu - \alpha\sigma^2)^2)}{2\sigma^2}$$

$$\frac{-[(x - (\mu - \alpha\sigma^2))^2 + \mu^2 - (\mu - \alpha\sigma^2)^2)]}{2\sigma^2}$$

Substituting this back into our original equation:

$$\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} exp\left(\frac{-[x - (\mu - \alpha\sigma^2) + \mu^2 - (\mu - \alpha\sigma^2)^2]}{2\sigma^2}\right)dx = 1$$

$$\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} exp\left(\frac{-[x - (\mu - \alpha\sigma^2)^2]}{2\sigma^2}\right)exp\left(\frac{-[\mu^2 - (\mu - \alpha\sigma^2)^2]}{2\sigma^2}\right)dx = 1$$

As the second exponential term of this equation contains no x variables, we can move it outside the integral and we are left with:

$$exp\left(\frac{-[\mu^2 - (\mu - \alpha\sigma^2)^2]}{2\sigma^2}\right)\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} exp\left(\frac{-[x - (\mu - \alpha\sigma^2)^2]}{2\sigma^2}\right)dx = 1$$

We also have the following:

$$\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} exp\left(\frac{-[x - (\mu - \alpha\sigma^2)^2]}{2\sigma^2}\right)dx = 1$$

This is true because this equation is the integral of the density function of a different normal distribution, where the mean is $\mu - \alpha\sigma^2$ instead of simply μ.

So we are left with:

$$exp\left(\frac{-[\mu^2 - (\mu - \alpha\sigma^2)^2]}{2\sigma^2}\right) = 1$$

Taking the natural logarithm of both sides:

$$(-\mu^2 - (\mu - \alpha\sigma^2)^2)/2\sigma^2 = 0$$

$$(-\mu^2 - (\mu - \alpha\sigma^2)^2) = 0$$

$$(\mu - \alpha\sigma^2)^2) = \mu^2$$

$$\pm\mu = \mu - \alpha\sigma^2$$

$0 = a$ OR $2\mu/\sigma^2 = a$

So we again have our familiar two solutions for a. One is where $a = 0$, and the other (the critical value for this case), is where $a = 2\mu/\sigma^2$.

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Substituting this back into our generic risk of ruin formula, we have:

$$R(b) = exp\left(\frac{-2\mu b}{\sigma^2}\right) \qquad \textbf{(22.4)}$$

This important result allows us to solve risk of rain problems for games with normal distributions. However, poker results are not in and of themselves normally distributed; in order to attempt to apply this risk of rain formula to poker, we must recognize the limitations of this approach. Poker games are not precisely normal. In general, poker hand results are clustered around particular outcomes, such as losing 1 small bet, losing 2 small bets, and so on. However, we know that the Central Limit Theorem allows us to conclude that the distribution of samples of sufficient size will converge on normal. The term "sufficient size" begs for clarification. Our ability to use the risk of rain formula in the previous section is dependent upon the size of sample that allows us to assume normality.

If we have a distribution (of any shape) **X** for a game and we have samples of a certain size such that the samples are roughly normal, we can calculate the risk of ruin of a game using the sampling distribution **X'**. This will provide an approximation of the risk of ruin of the game: however, this risk of ruin is often biased because of the difference between the sampling distribution and the underlying distribution.

One of the primary sources of bias here is ***skew.*** Skew is a cousin of variance. When we looked at the variance of a distribution, we considered the distance of each outcome from the mean and squared it. Tins resulted in positive numbers for each possible outcome, which were summed to find the variance of the distribution. Skew instead sums the cubes of the distances, which causes the values to have sign. Distributions where the weighted sum of the cubes of the distances is positive arc called skew positive and those where the sum is negative are called skew negative. When calculating skew, we also divide the sum of the cubed dinstances by the standard deviation of the distribution cubed.

The normal distribution has zero skew; it is symmetrical around the mean. By contrast, tournament distributions are highly skew positive. It is easy to see this by considering a 100-player, winner-take-all tournament. A player of average ability in this tournament would win 99 buyins 1% of the time and lose 1 buyin the other 99%. The ***expectation*** of this tournament would be zero, the ***variance*** would be 99 buyins^2/tourney, and the ***skew*** of the tourney would be about +9.84. Skew effectively measures the tendency of the distribution to have outliers or larger values to one side of the mean or another.

Distributions that have a lot of skew cause problems for the normal approximation method of calculating risk of ruin. We can attempt to calculate the risk of ruin for this tournament in two ways: first by using the normal approximation, and second by calculating *a* directly. Since risk of ruin is only directly applicable to games with positive expectation, we will consider a flayer with a full buyin overlay; that is, a player who wins 2% of the tournaments (as described above) that he enters.

First, the normal approximation:

w = 1 buyin/tourney
σ = 14.036 buyins/tourney
$R(b) = e^{-2b/197}$

For a convenient value of 100 buyins, we have a risk of ruin of about 36.2%.
Contrast this with the results we get from solving for *a* directly:

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$<e^{-ax}> = 1$
$^1/_{50}\,(e^{-99a}) + {}^{49}/_{50}\,(e^{a}) = 1$

Solving this equation by numerical methods, we get $a \approx 0.01683$.

Using this value in our RoR formula:

$R(100) = e^{-1.683} = 18.6\%$

So by solving directly for the risk of ruin of the game, we obtain a much lower risk of ruin. This is because the distribution of outcomes for a tournament has so much positive skew. It is possible to win 99 buyins m a single tournament; however at most we can lose 1 buyin. The normal approximation assumes that there will be a substantial number of larger negative outcomes, such as -5 or -10 buyins. However, the fact that these outcomes do not exist causes the normally distributed risk of ruin formula to overestimate the chance of initial ruin.

For this reason, we do not advocate the utilization of the normally distributed risk of ruin formula for tournaments. This, admittedly, was an extreme example because of the winner-take-all format, but the bias for skew right distributions is significant and present even for more graduated payout structures.

However, when we consider limit poker, we have a different situation. Here our initial distribution, though it is still somewhat skew right, is not nearly as distorted by this skew as the tournament example above.

As a starting point, we take the following distribution of hand outcomes, which, while simplified, is not too far away from what we might find for a typical mildly successful player.

Result in small bets (SB)	**Probability**	**Weighted contribution**
0	70%	0
-1	4.07%	-0.0407
-2	8.14%	-0.1627
-4	4.41%	-0.1763
-6	2.03%	-0.1220
-8	1.01%	-0.0814
-12	0.34%	-0.0407
+2	2.81%	0.0563
+4	3.44%	0.1375
+8	2.5%	0.2
+16	0.94%	0.15
+32	0.31%	0.1
Total	**100%**	**0.02 SB/hand**

If the reader disagrees with this characterization of the shape of the distribution of hand outcomes, the particular numbers chosen are not essential to the analysis that follows.

This distribution has a mean of 0.02 SB/hand and a variance of 10.80 SB^2/hand. Using the normal approximation for a bankroll of 300 small bets, we find the risk of ruin to be:

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$e^{-(2)(300)(0.02)/10.80} = 32.89\%$

We can set up Equation 22.1 for this distribution as well. This equation is fairly complex; we omit the details, although it is no different than our previous problems, only with more numerous terms added together.

Solving this with numerical methods, we find a = 0.00378. The risk of ruin of a 300-unit bankroll with this a value is:

$e^{-300a} = 32.17\%$

As you can see, these values are quite close; this property holds for most typical distributions of hand results. The point of this is that even though the skew of these distributions does exist, it is nowhere near as large as the skew of tournament outcomes (this remains the case even when we have graduated payout structures, as we shall see in a moment). Distributions of hand results and the normal distribution appear, in our experience, and from the data we've worked with, to be close enough to justify using the normal distribution risk of ruin as an accurate approximation to the full risk of ruin formula.

To review what we have covered so far. First we looked at tournaments. The primary characteristic of interest with regard to tournament distributions is that they are significantly skew right. When we apply the normal distribution risk of ruin function to tournament mean and variance, we overstate the risk of ruin significantly. This occurs because we lack large negatives; each tournament can only cost a single buyin total. Second, we considered limit ring games. Here we found that the distribution was sufficiently normal that the normal distribution risk of ruin formula was relatively accurate. So the characteristics of most importance in assessing risk of ruin for limit ring games are mean and variance.

A third type of distribution is common and increasing in popularity. This is the no-limit ring game distribution. Here we have large positives from hands where all the money goes in and we win, but also large negatives from hands where we lose our entire stacks. As a result of this, we find that the normal distribution risk of ruin again is a biased approximation. The characteristic of most importance here is ***kurtosis,*** which is the tendency of the tails of the distribution to be "fat," or the tendency of outcomes far from the mean to occur more often than the normal distribution would indicate. Recall that, for example, events three standard deviations away from the mean of the normal distribution occur only 0.3% of the time. However, in a game like no-limit, events several standard deviations away happen with a higher frequency. As it turns out, this effect is fairly minimal in its effect on risk of ruin for most distributions. Consider the following highly simplified distribution:

Result (in big blinds)	**Probability**	**Weighted contribution**
0	70%	0
+1	15%	0.15
-1	13%	-0.13
+50	1.1%	0.55
-50	0.9%	-0,45
Total	**100%**	**0.12 BB/hand**

The point here is that we can compare the risk of ruin values computed directly and from the normal distribution risk of ruin formula and see that the normal distribution understates the risk of ruin. Suppose our bankroll for this game is 500 units.

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The mean of this distribution is 0.12 BB/hand, and the variance is 50.27 BB^2/hand.

$R(500) = e^{-2(0.12)(500)/50.27} = 9.19\%$

Solving Equation 22.1 for this distribution, we find that

$a = 0.004789$

The risk from this value is:

$e^{-0.004789(500)} = 9.12\%$

So in this particular distribution, the kurtosis has very little effect. In fact, the risk of ruin is lower when calculated directly than by using the normal distribution. As a result of this and other calculations, it seems that the normal distribution provides an approximation with tolerable error that can be used in assessing risk of ruin in no-limit ring games as well.

However, it bears noting that no-limit ring game results can also be skew right in the same manner as tournaments, particularly for players who play very rightly in big pots. If a player loses many small pots but wins most of the big ones he enters due to playing style, the kurtosis has minimal effect, but the skew will tend to cause the risk of ruin formula to overestimate the chance of ruin.

We summarize the findings of this section:

Type of Game	**Risk formula to use**	**Error of normal distribution**
Limit ring games	Normal Distribution	Small
Tournaments	Direct Calculation	Large (using normal distribution results is values that are too high)
No-limit ring games	Normal Distribution	Small

Key Concepts

- The risk of ruin model can give us valuable insights into our chance of going broke from a particular bankroll level. This model assumes that we will play a single game with a known distribution of outcomes either indefinitely or until we lose the starting stake. The risk of ruin *R(b)* is the chance that we will go broke starting from a bankroll b.
- Risk of ruin is 100% for games with negative expectation.
- Equation 22.2 provides us a method of calculating risk of ruin for any game with a fixed distribution of outcomes.
- Maintaining a bankroll at a given level will not provide insulation from risk of ruin; if we have a bankroll N and we take all the money above N out of the bankroll on a regular basis, our risk of ruin is 100%.
- Half-bankrolls are a useful shorthand for risk of ruin that can allow us to make off-the-cuff risk calculations by simply finding the number of half-bankrolls needed to assure a particular risk level.
- For normally distributed outcomes, Equation 22.4 gives the risk of ruin for a particular bankroll.
- For some distributions, such as tournaments, we cannot use the normal distribution formula because the outcome distributions are too skew. Instead, we can solve directly using Equation **22.2.**

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